

Background

Compact high-redshift sources are modelled by subtracting a point-spread function built from field stars. Inferred sizes, luminosities and masses all rest on that subtraction being colour-correct.

No test asked whether a stellar PSF stays valid when the target's spectrum differs from the stars'.

What this paper is selling

The authors derive why broad-band stellar PSFs systematically misassign flux for spectrally distinct compact sources, then demonstrate a correction. Applied to little red dots, the bias inflates or invents extended host components.

Key results to remember

- A pure point source yields wavelength-dependent residuals mimicking a Sérsic host.
- Every tested nuclear spectrum, stellar reference and detector position exceeds the published 11.8% F444W host.
- The bias is large enough to change inferred sizes, luminosities and masses.

